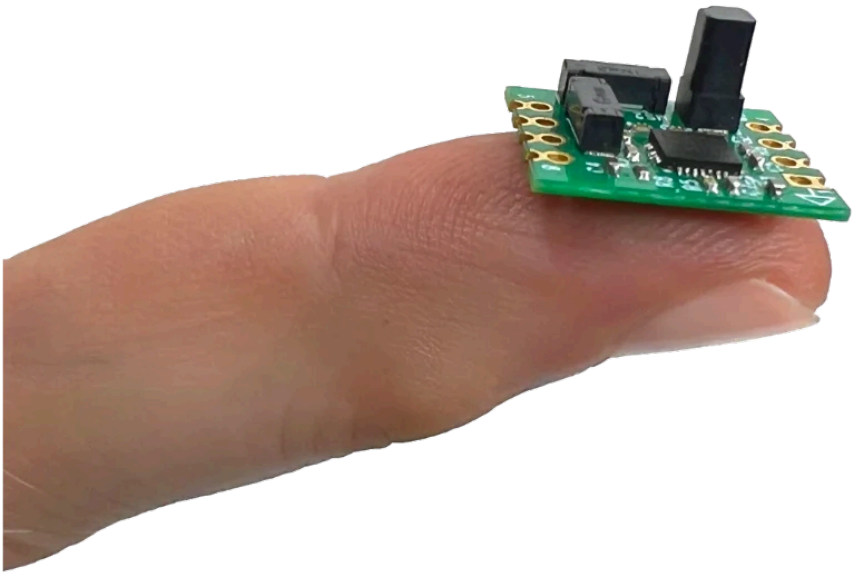
Ruggedized to withstand extreme vibration

RM3100-CB

Compact Board-Mounted Magnetometer

All the benefits of the RM3100 on a tiny, easy-to handle board for quick and simple installation.

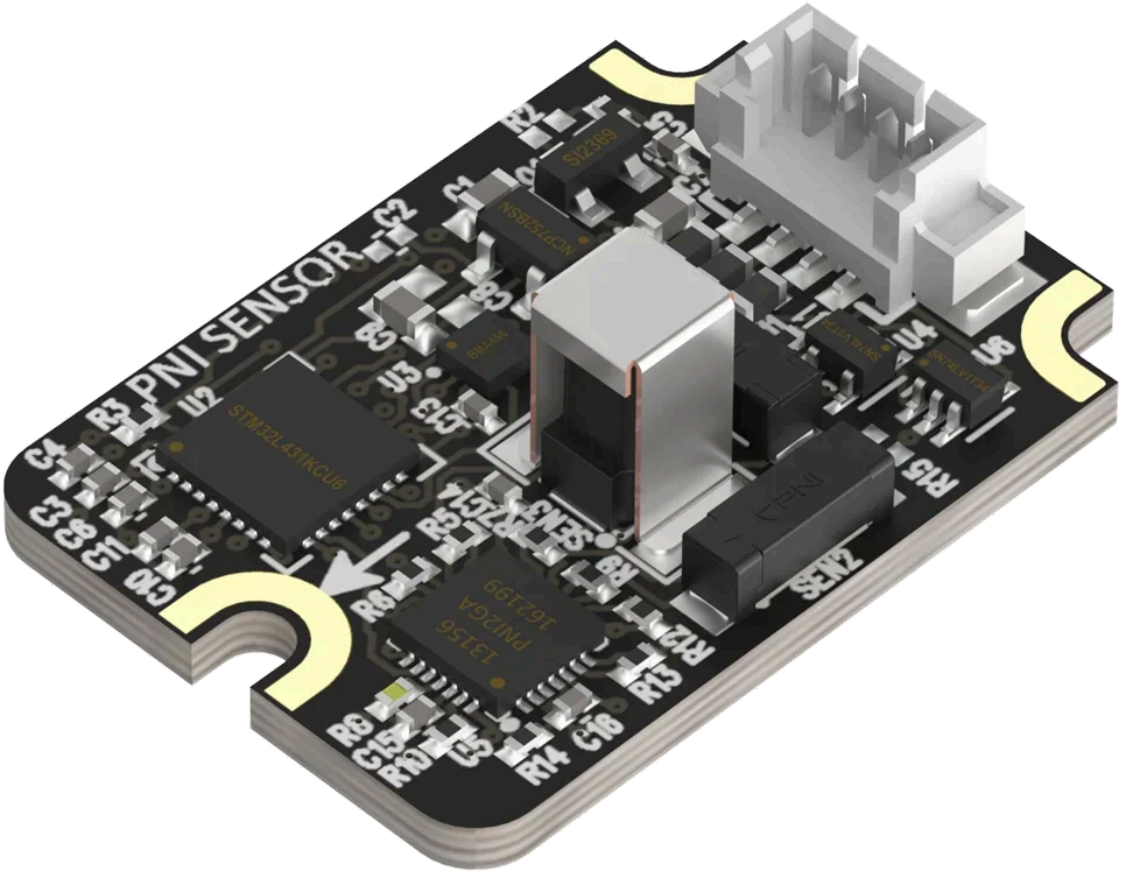


- Getting accurate navigation and position data depends on having accurate magnetic field measurement to begin with. Unlike other COTS (Commercial Off the Shelf) magnetic sensors, the RM3100-CB magnetometer has over 23 times better resolution and 33 times less noise resulting in far more precise data outputs.
- The RM3100-CB can deflect fields as low as 2.7 nT at 1Hz, with a noise floor of 4 pT /√Hz at 1Hz. It also has a linear response over a range of 100,000 nT. It's been tested for various environmental conditions, such as thermal shock, mechanical shock, radiation, and thermal cycles, and has shown strong performance and stability.
- Stable and reliable over time and temperature and extremely radiation tolerant, the RM3100-CB has I2C and SPI interfaces for system design flexibility and connects easily to Arduino, Raspberry Pi, and other microcontroller boards.
- Like the RM3100, the RM3100-CB delivers impeccable signal-to-noise, drift-free, precise magnetic field measurements, ideal for underwater, ground, air, and space applications. It offers Incredible resolution with low power, low mass, small size and large dynamic range. And because it's board- mounted into a single unit, it's quick and easy to install.
- RM3100-CB is ideal for use in situations which involve significant vibration including space launches. Using the MIL-STD-810H test method 514.8 for general transportation ruggedness, this magnetometer has demonstrated robustness against intense vibration at 50gs RMS.

Technical **Specifications**

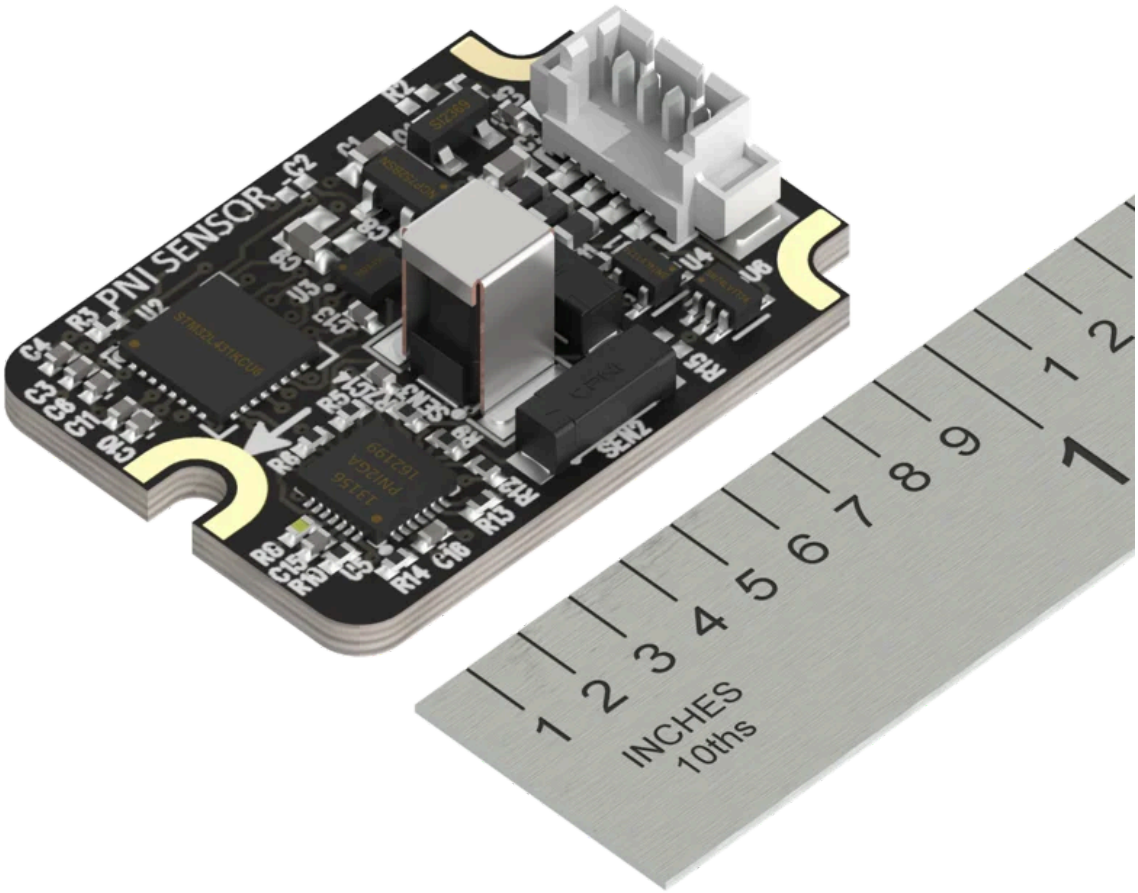
Sensitivity	13nT
Noise	15nT
Linearity over +/- 200 uT	0.50%
Field Measurement Range	+/- 1100 uT
Current @ 8Hz, 3 Axes	260 uA
Interface	SPI and I2C
Operating Temperature	-40°C to +85°C
Size (L x W x H) mm	14.22 x 15.75 x 7.1

Other PNI products you might be interested in:



Prime Pro ⇒

Digital Magnetic Compass



NaviGuider ⇒

IMU orientation module system for UAVs, robots and buoys

Ask **our experts**

Get direct advice from our engineers in less than 24 hours. We are available Monday through Friday, 9:00 am to 4:00 pm, Pacific Time.

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